LAYOUT PROPOSAL FOR THE LOGISTICS AND OPERATIONAL SUPPORT CENTER OF ETAPA EP

- Geovanny Patricio Guzmán Cordero
- Andrés Esteban Ortega Andrade
- **Tutor: Paola Vintimilla Álvarez**

Abstract

To determine and design a layout proposal for the logistics and operations center of ETAPA EP through the use of Industrial Engineering tools and simulation.

Objectives

Main Objective

To develop a layout proposal focused on space optimization, effective inventory management, and continuous improvement of operational processes.

Specific Objective

To collect information on inventory processes, as well as direct and indirect labor, in order to establish specific requirements for the layout.

Specific Objective

To identify the main indicators that allow evaluating the efficiency of a layout model.

Specific Objective

To design and simulate different layout scenarios using FlexSim.

To compare and evaluate the KPIs of each scenario in order to select the best proposal.

Methodology

Phase 1	Phase 2	Phase 3	Phase 4	Phase 5
Diagnosis and Data Collection	**Identification of Key Performance Indicators (KPIs).**	**Design of the layout model**	**Design simulation using FlexSim.**	**Evaluation and selection of the optimal scenario.**
Technical data collection of the current facility	Selection of relevant KPIs for the logistics center.	Application of the Systematic Layout Planning (SLP) methodology.	Modeling and validation of different scenarios.	Multi-criteria analysis of scenarios based on KPIs.

Curricular Integration

Industrial CAD Design.
Logistics and Supply Chain Management.
Process Engineering.
Manufacturing Process Simulation.

Applied Engineering

SLP Methodology.
Application of ABC-XYZ Analysis.
Process Mapping.
Discrete Event Simulation using FlexSim.

Keywords

Facility layout · Logistics center · Simulation · Optimization

Results

Inventory is highly concentrated: category A (177 items) accounts for 80% of consumption value (USD 295,139.4), while category C (1,209 items) represents only 5% of the value.

The capacity calculation shows a total storage requirement of 5,331.88 m², so vertical storage (4,001.64 m² equivalent) using shelving/racks is essential

In simulation, Proposal 2 improves speed and costs (e.g., total reference cost USD 79.50 vs USD 363.56) but increases physical workload (operators' distance 8,122.5 m vs 5,794.2 m); for sustainability/ergonomics and space utilization (67.43% vs 62.71%), Proposal 1 is recommended as the baseline.

Conclusions

Conclusión 1

Family-based classification and ABC–XYZ analysis allowed for a functional inventory structure, providing a solid foundation for space dimensioning.

Conclusión 2

The detailed item-by-item space calculation—including the definition of shelving, racks, and floor storage areas—enabled the design of two warehouse layout proposals.

Conclusión 3

Both proposals were aligned with turnover patterns, prioritizing the proximity of critical items to the picking point.

Conclusión 4

The results demonstrate that there is no absolute superiority of one proposal over the other; instead, they show differentiated performance based on the evaluated indicators and institutional strategic priorities.

Contact Information

•Geovanny Patricio Guzmán Cordero
patricio.guzman@ucuenca.edu.ec

Andrés Esteban Ortega Andrade
esteban.ortega@ucuenca.edu.ec

References

Díaz Martínez, M. A., Zárate Cruz, R., & Román Salina, R. V. (2018).
Moura, C. R., Borges, W. J., & Meincheim, E. (2021).
Ortiz Naranjo, E. J., & Zúñiga Valle, A. X. (2022).